Foundations of Programming

Assignment No.: 19

PROBLEM STATEMENT: Create Structure EMPLOYEE for storing details (Name, Designation, gender, Date of Joining and Salary). Define function members to compute a) total number of employees in an organization b) count of male and female employee c) Employee with salary more than 10,000 d) Employee with designation “Asst Manager”

OBJECTIVE:

- To understand the use of structures in C programming.
- To store and manage employee records using structures.
- To perform operations such as counting and filtering employee data

THEORY:

In the C programming language, a structure is a user-defined data type that groups related variables of different data types under a single name. Structures are widely used for representing records such as employee information. In this program, a structure named EMPLOYEE is created to store employee details such as name, designation, gender, date of joining, and salary. By using loops and conditional statements, the program can calculate the total number of employees, count male and female employees, find employees whose salary is greater than 10,000, and display employees having the designation 'Asst Manager'.

PLATFORM: Linux

ALGORITHM:

1. Star
2. Define structure EMPLOYEE with fields:
- Name
- Designation
- Gender
- Date of Joining
- Salary
3. Input number of employees n
4. Create an array of EMPLOYEE of size n
5. Input details for all employees
6. Total number of employees:
- Total = n
- Display total
7. Count male and female employees:
- Set male = 0, female = 0
- For each employee:
  - If gender = male → male++
  - Else if gender = female → female++
- Display counts
8. Employees with salary > 10000:
- For each employee:
  - If salary > 10000:
    - Display employee details
9. Employees with designation “Asst Manager”:
- For each employee:
  - If designation = “Asst Manager”:
    - Display employee details

10. Stop

INPUT:

```
 #include <stdio.h>
#include <string.h>
struct EMPLOYEE {
    char name[50];
    char designation[50];
    char gender;
    char doj[15]; // Date of Joining
    float salary;
};
void processEmployeeData(struct EMPLOYEE emp[], int n) {
    int male_counter = 0;
    int female_counter = 0;
    int asst_manage_counter = 0;
    int i;
    printf("\n--- Processing Results ---\n");
    printf("Employees with salary > 10,000:\n");
    for (i = 0; i < n; i++) {
        if (emp[i].gender == 'M' || emp[i].gender == 'm') {
            male_counter++;
        } else if (emp[i].gender == 'F' || emp[i].gender == 'f') {
            female_counter++;
        }
        if (emp[i].salary > 10000) {
            printf("- %s\n", emp[i].name);
        }
        if (strcmp(emp[i].designation, "Asst Manager") == 0) {
            asst_manage_counter++;
        }
    }
    printf("\nTotal Employees: %d\n", n);
    printf("Male Employees: %d\n", male_counter);
    printf("Female Employees: %d\n", female_counter);
    printf("Assistant Managers: %d\n", asst_manage_counter);
}
int main() {
    int n, i;
    printf("Enter number of employees (n): ");
    scanf("%d", &n);
```

```
    struct EMPLOYEE emp[n];
    for (i = 0; i < n; i++) {
        printf("\n--- Enter details for Employee %d ---\n", i + 1);
        printf("Name: ");
        scanf("%s", emp[i].name);
        printf("Designation (e.g., Asst Manager): ");
        scanf(" %[^\n]", emp[i].designation); // Reads string with spaces
        printf("Gender (M/F): ");
        scanf(" %c", &emp[i].gender);
        printf("Date of Joining (DD/MM/YYYY): ");
        scanf("%s", emp[i].doj);
        printf("Salary: ");
        scanf("%f", &emp[i].salary);
    }
    processEmployeeData(emp, n);
    return 0;
}
```

```c
#include <stdio.h>
#include <string.h>

struct EMPLOYEE {
    char name[50];
    char designation[50];
    char gender;
    char doj[15]; // Date of Joining
    float salary;
};

void processEmployeeData(struct EMPLOYEE emp[], int n) {
    int male_counter = 0;
    int female_counter = 0;
    int asst_manage_counter = 0;
    int i;

    printf("\n--- Processing Results ---\n");
    printf("Employees with salary > 10,000:\n");

    for (i = 0; i < n; i++) {

        if (emp[i].gender == 'M' || emp[i].gender == 'm') {
            male_counter++;
        } else if (emp[i].gender == 'F' || emp[i].gender == 'f') {
            female_counter++;
        }

        if (emp[i].salary > 10000) {
            printf("- %s\n", emp[i].name);
        }

        if (strcmp(emp[i].designation, "Asst Manager") == 0) {
            asst_manage_counter++;
        }
    }

    printf("\nTotal Employees: %d\n", n);
    printf("Male Employees: %d\n", male_counter);
    printf("Female Employees: %d\n", female_counter);
    printf("Assistant Managers: %d\n", asst_manage_counter);
}

int main() {
    int n, i;

    printf("Enter number of employees (n): ");
    scanf("%d", &n);

    struct EMPLOYEE emp[n];

    for (i = 0; i < n; i++) {
        printf("\n--- Enter details for Employee %d ---\n", i + 1);
        printf("Name: ");
        scanf("%s", emp[i].name);
        printf("Designation (e.g., Asst Manager): ");
        scanf(" %[^\n]", emp[i].designation); // Reads string with spaces
        printf("Gender (M/F): ");
        scanf(" %c", &emp[i].gender);
        printf("Date of Joining (DD/MM/YYYY): ");
        scanf("%s", emp[i].doj);
        printf("Salary: ");
        scanf("%f", &emp[i].salary);
    }

    processEmployeeData(emp, n);

    return 0;
}
```

OUTPUT:

Enter number of employees: 3

Employee 1 details:

Name: Rahul

Designation: Manager

Gender (Male/Female): Male

Date of Joining (dd-mm-yyyy): 01-01-2020

Salary: 15000

Employee 2 details:

Name: Sneha

Designation: AsstManager

Gender (Male/Female): Female

Date of Joining (dd-mm-yyyy): 15-03-2021

Salary: 12000

Employee 3 details:

Name: Amit

Designation: Clerk

Gender (Male/Female): Male

Date of Joining (dd-mm-yyyy): 10-07-2022

Salary: 8000

Total number of employees = 3

Male employees = 2

Female employees = 1

Employees with salary more than 10000: Rahul Sneha

Employees with designation AsstManager: Sneha

CONCLUSION: Thus, the C program successfully demonstrates the use of structures to store employee details and perform various operations such as counting employees, identifying employees with salary greater than 10,000, and displaying employees with the designation 'Asst Manager'.

FAQs:

1. How does a structure in the C programming language help in organizing multiple related data items such as employee details?
2. Why are structures preferred over simple variables or arrays when managing records like employee information?

3. How can conditional statements and functions be used with structures to filter employees based on salary or designation?